

Top Entry Ball Valve

Structural Chracteristic

We use the Germany advanced technology and apply it in our new products by years of R&D, we develop Wanli special Top-entry ball valve. With technology up to domestic leading level, these products can compete with the international counterparts. Besides the features borne by the side installed fixed ball valve (see fixed ball valve), they are also given the features below:

1. One-piece valve body, top installed trunnion to fix and support, flanged or butt welded structure.
2. Inconel bellows spring makes the sealed metal seat moving toward the ball, providing the two way seal function of inlet and outlet.
3. Distinctive technology of retractable valve seat, inline replace ability of ball arm, valve stem sealing ring, metal seat and bellows spring, and torque down to the minimum for the ease of operation.
4. Ease of repair and installation, long performance life. Access for repair and replacement of trims may be done with no need to remove the valve from the pipeline, thus to lengthen the service life.
5. Low torque, reliable sealing, and spherical seat different from ordinary ball valves, which can automatically adjust the position of seal.
6. Preset platform and bolt holes for drive device, and to ISO5211. Drive device can be installed according to users' different requirements.
7. DBB (double block and bleed) function.
8. Hand operated top installed ball valve adopts high strength one-piece ball arm structure to ensure the precise positioning of the ball.
9. Anti-blowout protective structure of valve stem to enhance the operational safety of the valve.

Ball Valve Technical Specifications

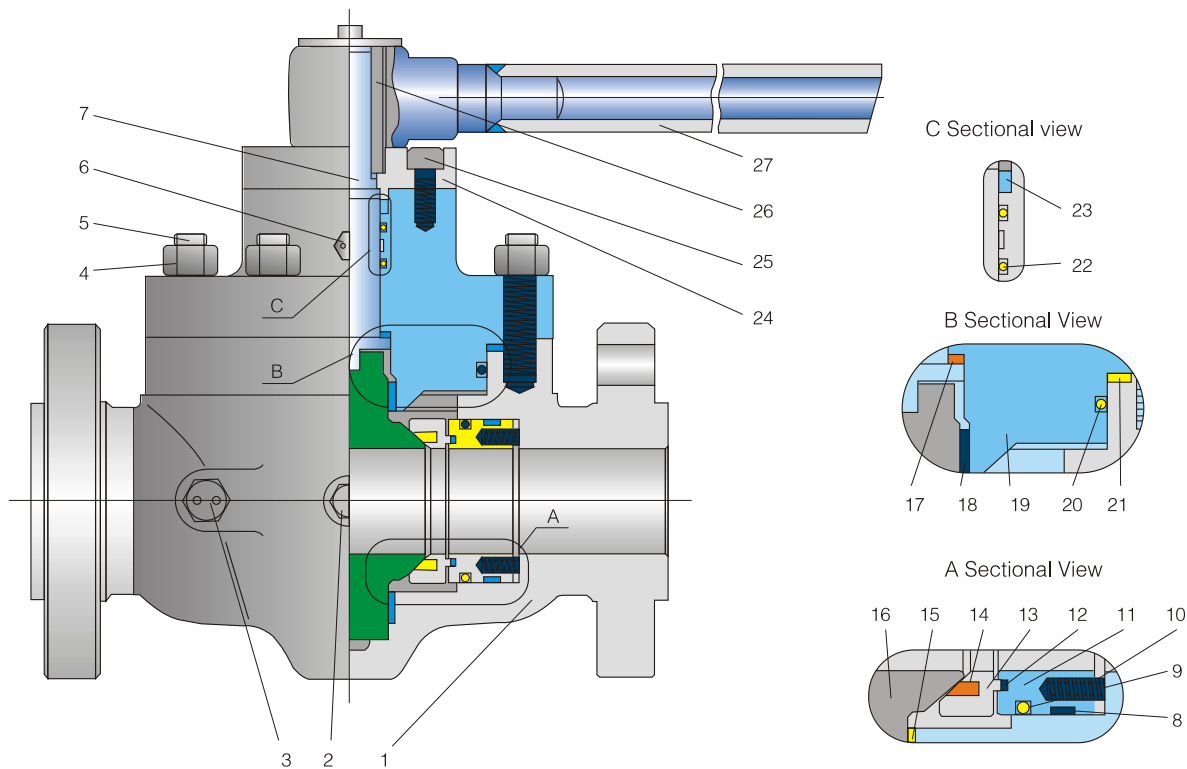
Technical Specifications	API Series
Design Specifications	API6D、API608、BS5351
Pressure–Temperture Ratings	ASME B16.34
Face–to–Face	API 6D、ASME B16.10
Flanged Ends	ASME B16.5、ASME B16.47
Butt–welding Ends	ASME B16.25
Inspection and Test	API6D、API598
Fire–safe Test	API 607、API 6FA
Castings Quality Inspection	MSS–SP–55

Performance Specification

Performance Specification		Pressure Class						
		150	300	400	600	900	1500	2500
Test Pressure (MPa)	Strength Test	2.94	7.67	10.20	15.30	22.98	38.30	63.83
	Sealing Test	2.16	5.62	7.48	11.23	16.85	28.08	46.81
	Air Test	0.6						
Applicable Temperature		-196℃ ~ 550℃ (Different materials for different working conditions.)						
Applicable Medium		Water, Steam, Oil, Gas, Liquid Gas, Natural Gas etc						
Size Range		DN50 ~ 900(NP2" ~ 36"), We can produce according to your special requirements.						
Body/Trim Material		Carbon Steel, Stainless Steel, Duplex Stainless Steel, Ni-Alloy Steel, Ti Material						
Ends Connection		Flange Connection,Butt-welded Connection						
Actuating Device		Lever, Gearbox, Electric, Pneumatic etc.						

Top Entry Ball Valve

Valve Structural Schematic Drawing (Manual Top Entry Ball Valve)



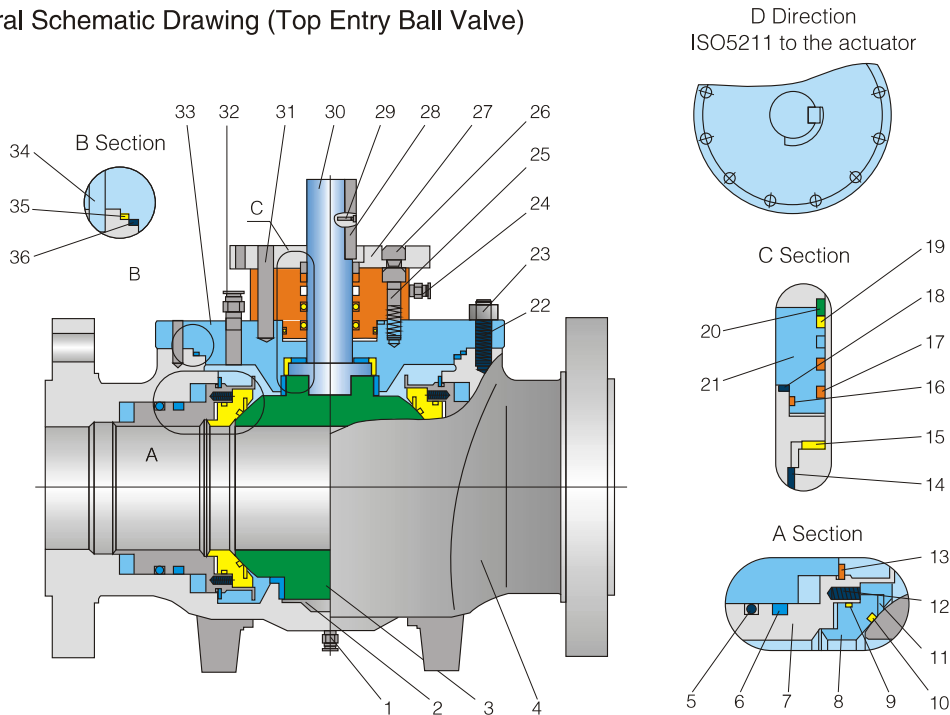
The Material of Main Parts

No.	Parts Name	Material		
		Carbon Steel	SS	Low Temperature Steel
1	Body	A216 WCB	A351 CF8M	A352 LCC
2	Drain Valve	A29 1025	SS	SS
3	Sealant Injection Valve	A29 1025	SS	SS
4	Nut	A194 2H	A194 8	A194 7
5	Stud	A193 B7	A193 B8	A320 L7
6	Sealant Injection Valve	A29 1025	SS	SS
7	Stem	A182 F6a	A182 F316	A182 F316
8	Fire Safe Ring	Soft Graphite		
9	O-Ring	VITON		
10	Spring	INCONEL X-750		
11	Supporting Ring	A29 4140+ENP	A182 F316	A350 LF3
12	Gasket	Soft Graphite+SS		
13	Seat	A105+ENP A29 4140+ENP	A182 F316	A350 LF3
14	Sealing Ring	PTFE、NYLON、PEEK、PCTFE、MOLON		
15	Bushing	PTFE+CS	PTFE+SS	PTFE+SS
16	Ball	A105+ENP	A182 F316	A350 LF3
17	Flat Bushing	PTFE+CS	PTFE+SS	PTFE+SS
18	Bushing	PTFE+CS	PTFE+SS	PTFE+SS
19	Bonnet	A216 WCB	A351 CF8M	A352 LCC
20	O-Ring	VITON		
21	Gasket	Soft Graphite+SS		
22	O-Ring	VITON		
23	Packing	Soft Graphite		
24	Cover	A29 4140+ENP	A182 F316	A350 LF3
25	Screw	A193 B7	A193 B8	A320 L7
26	Key	A29 1045	A29 1045	A29 1045
27	Lever	Q235A		

Notes: 1. Materials are available according to users' requirements and different working medium
2. Materials for Anti-sulfur valve shall be chosen according to NACE MR0175.

Top Entry Ball Valve

Valve Structural Schematic Drawing (Top Entry Ball Valve)



The Material of Main Parts

No.	Parts Name	Material		
		Carbon Steel	SS	Low Temperature Steel
1	Drain Valve	A29 1025	SS	SS
2	Bushing	PTFE+CS	PTFE+SS	PTFE+SS
3	Ball	A105+ENP	A182 F316	A350 LF3
4	Stem	A216 WCB	A351 CF8M	A352 LCC
5	O-Ring	VITON		
6	Fire Safe Ring	Soft Graphite		
7	Supporting Ring	A29 4140+ENP	A182 F316	A350 LF3
8	Seat	A105+ENP	A182 F316	A350 LF3
9	O-Ring	VITON		
10	O-Ring	VITON		
11	Sealing Ring	PTFE、NYLON、PCTFE、MOLON		
12	Spring	INCONEL X-750		
13	C Type Spring	17-4		
14	Bushing	PTFE+CS	PTFE+SS	PTFE+SS
15	Flat Bush Bearing	PTFE+CS	PTFE+SS	PTFE+SS
16	O-Ring	VITON		
17	O-Ring	VITON		
18	Gasket	Soft Graphite+SS		

No.	Parts Name	Material		
		Carbon Steel	SS	Low Temperature Steel
19	Packing	Soft Graphite		
20	Gland	A182 F6a	A182 F6a	A350 LF3
21	Cover	A29 4140+ENP	A182 F6a	A320 L7
22	Stud	A193 B7	A193 B8	A194 7
23	Nut	A194 2H	A194 8	A350 LF3
24	Sealant Injection Valve	A29 1025	SS	SS
25	Screw	A193 B7	A193 B8	A320 L7
26	Screw	A193 B7	A193 B8	
27	Connecting Plate	A105		
28	Key	A29 1045		
29	Screw	A193 B7		
30	Stem	A182 F6a	A182 F316	A182 F316
31	Locating Pin	A182 F6a		
32	Sealant Injection Valve	A29 1025	SS	SS
33	Bonnet	A216 WCB	A351 CF8M	A352 LCC
34	Locating Pin	A182 F6a		
35	Gasket	Soft Graphite+SS		
36	O-Ring	VITON		

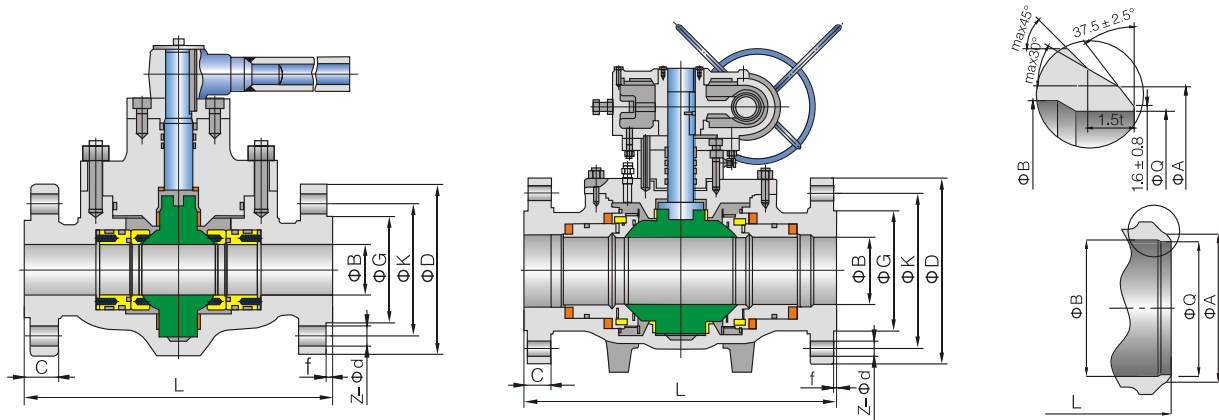
Note: 1. Materials are chosen according to the temperature & pressure for different working mediums.

2. It can be available to choose the materials according to user's requirements, besides the listed material in the chart.

3. Materials in accordance with NACE MR-01-75(latest edition) are available for sour gas service.

Remarks: RF stands for Raised Flange Connection; BW stands for Buttwelded Ends Connection; RTJ stands for Ring Joint Connection.

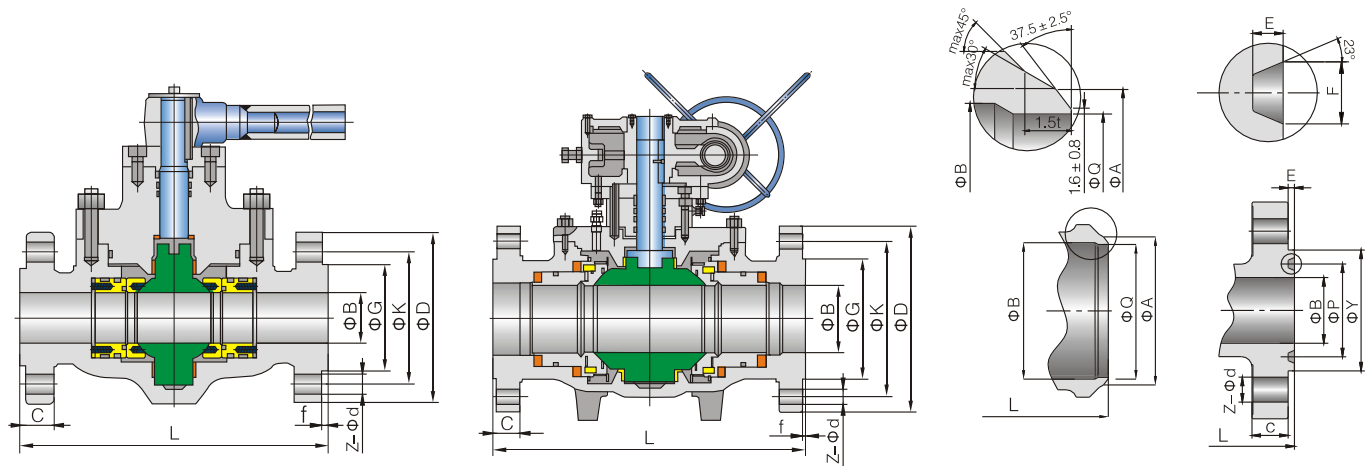
Full Port Top Entry Ball Valve



Main Connection Dimension

Class	Size		Dimensions(mm)										
	NPS	DN	L		B	D	G	f	K	Z-Φd	C	A	Q
			RF	BW									
150Lb	2	50	178	216	49	152	91.9	1.6	120.7	4-19.1	15.8	63	Asper client's requirement
	2 1/2	65	191	241	62	178	104.6	1.6	139.7	4-19.1	17.5	75	
	3	80	203	283	74	191	127	1.6	152.4	4-19.1	19.1	91	
	4	100	229	305	100	229	157.2	1.6	190.5	8-19.1	23.9	117	
	6	150	394	457	150	279	215.9	1.6	241.3	8-22.2	25.4	172	
	8	200	457	521	201	343	269.7	1.6	298.5	8-22.2	28.4	223	
	10	250	533	559	252	406	323.9	1.6	362	12-25.4	30.2	278	
	12	300	610	635	303	483	381	1.6	431.8	12-25.4	31.8	329	
	14	350	686	762	334	533	412.8	1.6	476.2	12-28.6	35.1	362	
	16	400	762	838	385	597	469.9	1.6	539.8	16-28.6	36.6	413	
	18	450	864	914	436	635	533.4	1.6	577.9	16-31.8	39.6	464	
	20	500	914	991	487	699	584.2	1.6	635	20-31.8	42.9	516	
	24	600	1067	1143	589	813	692.2	1.6	749.3	20-34.9	47.8	619	
	26	650	1143	1245	633	870	749.3	1.6	806.5	24-34.9	68.3	670	
	28	700	1245	1346	684	927	800.1	1.6	863.6	28-34.9	71.4	721	
	30	750	1295	1397	735	984	857.3	1.6	914.4	28-34.9	74.7	772	
	32	800	1372	1524	779	1060	914.4	1.6	977.9	28-41.3	80.8	825	
	34	850	1473	1626	830	1111	965.2	1.6	1028.7	32-41.3	82.6	876	
	36	900	1524	1727	874	1168	1022.4	1.6	1085.9	32-41.3	90.4	927	

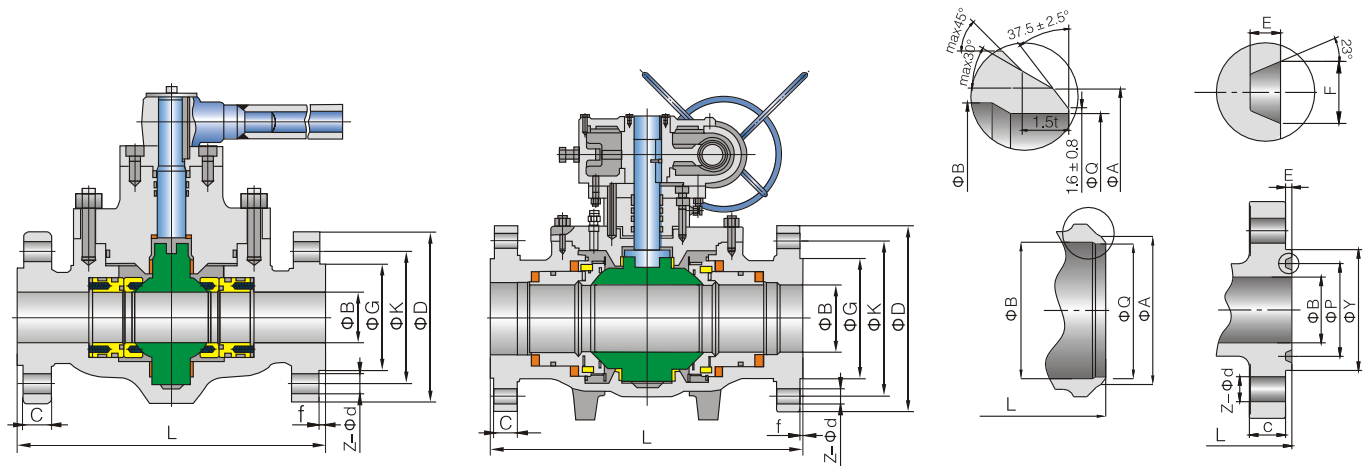
Full Port Top Entry Ball Valve



Main Connection Dimension

Class	Size		Dimensions(mm)															Q
	NPS	DN	L			B	D	G	f	K	Z-Φd	C	Groove				A	
			RF	RTJ	BW								Y	P	E	F		
300Lb	2	50	216	232	216	49	165	91.9	1.6	127	8-19.1	22.4	108	82.55	7.92	11.91	63	Asper client's requirement
	2 1/2	65	241	257	241	62	191	104.6	1.6	149.2	8-22.2	25.4	127	101.6	7.92	11.91	75	
	3	80	283	298	283	74	210	127	1.6	168.3	8-22.2	28.4	146	123.82	7.92	11.91	91	
	4	100	305	321	305	100	254	157.2	1.6	200.2	8-22.2	31.8	175	149.22	7.92	11.91	117	
	6	150	403	419	457	150	318	215.9	1.6	269.8	12-22.2	36.6	241	211.12	7.92	11.91	172	
	8	200	502	518	521	201	381	269.7	1.6	330.2	12-25.4	41.1	302	269.88	7.92	11.91	223	
	10	250	568	584	559	252	445	323.9	1.6	387.4	16-28.6	47.8	356	323.85	7.92	11.91	278	
	12	300	648	664	635	303	521	381	1.6	450.9	16-31.8	50.8	413	381	7.92	11.91	329	
	14	350	762	787	762	334	584	412.8	1.6	514.4	20-31.8	53.8	457	419.1	7.92	11.91	362	
	16	400	838	854	838	385	648	469.9	1.6	571.5	20-34.9	57.2	508	469.9	7.92	11.91	413	
	18	450	914	930	914	436	711	533.4	1.6	628.7	24-34.9	60.5	575	533.4	7.92	11.91	464	
	20	500	991	1010	991	487	775	584.2	1.6	685.8	24-34.9	63.5	635	584.2	9.52	13.49	516	
	24	600	1143	1165	1143	589	914	692.2	1.6	812.8	24-41.3	69.9	749	692.15	11.13	16.66	619	
	26	650	1245	1270	1245	633	972	749.3	1.6	876.3	28-44.5	79.2	809.8	749.3	12.7	19.84	670	
	28	700	1346	1372	1346	684	1035	800.1	1.6	939.8	28-44.5	85.9	860.6	800.1	12.7	19.84	721	
	30	750	1397	1422	1397	735	1092	857.3	1.6	997	28-47.6	91.9	917.4	857.25	12.7	19.84	772	
	32	800	1524	1533	1524	779	1149	914.4	1.6	1054.1	28-50.8	98.6	984.3	914.4	14.27	23.01	825	
	34	850	1626	1654	1626	830	1207	965.2	1.6	1104.9	28-50.8	101.6	1035.1	965.2	14.27	23.01	876	
	36	900	1727	1756	1727	874	1270	1022.4	1.6	1168.4	32-54	104.6	1092.2	1022.35	14.27	23.01	927	

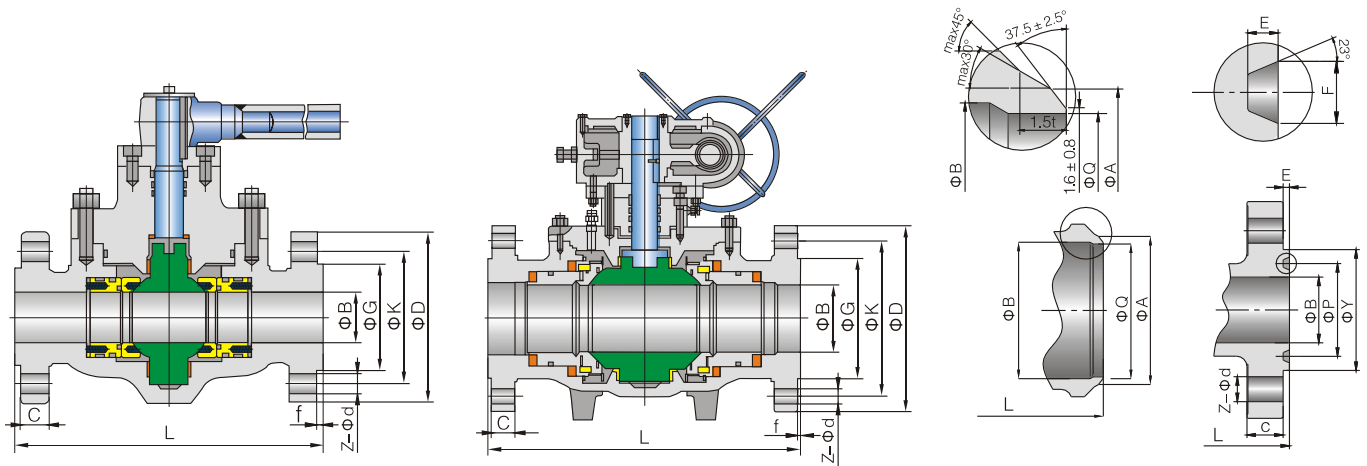
Full Port Top Entry Ball Valve



Main Connection Dimension

Class	Size		Dimensions(mm)															Q
	NPS	DN	L			B	D	G	f	K	Z-Φd	C	Groove				A	
			RF	RTJ	BW								Y	P	E	F		
600Lb	2	50	292	295	292	49	165	91.9	6.4	127	8-19.1	25.4	108	82.55	7.92	11.91	63	Asper client's requirement
	2 1/2	65	330	333	330	62	191	104.6	6.4	149.2	8-22.2	28.4	127	101.6	7.92	11.91	75	
	3	80	356	359	356	74	210	127	6.4	168.3	8-22.2	31.8	146	123.82	7.92	11.91	91	
	4	100	432	435	432	100	273	157.2	6.4	215.9	8-25.4	38.1	175	149.22	7.92	11.91	117	
	6	150	559	562	559	150	356	215.9	6.4	292.1	12-28.6	47.8	241	211.12	7.92	11.91	172	
	8	200	660	664	660	201	419	269.7	6.4	349.2	12-31.8	55.6	302	269.88	7.92	11.91	223	
	10	250	787	791	787	252	508	323.9	6.4	431.8	16-34.9	63.5	356	323.85	7.92	11.91	278	
	12	300	838	841	838	303	559	381	6.4	489	20-34.9	66.5	413	381	7.92	11.91	329	
	14	350	889	892	889	334	603	412.8	6.4	527.1	20-38.1	69.9	457	419.1	7.92	11.91	362	
	16	400	991	994	991	385	686	469.9	6.4	603.2	20-41.3	76.2	508	469.9	7.92	11.91	413	
	18	450	1092	1095	1092	436	743	533.4	6.4	654.1	20-44.5	82.6	575	533.4	7.92	11.91	464	
	20	500	1194	1200	1194	487	813	584.2	6.4	723.9	24-44.5	88.9	635	584.2	9.52	13.49	516	
	24	600	1397	1407	1397	589	940	692.2	6.4	838.2	24-50.8	101.6	749	692.15	11.13	16.66	619	
	26	650	1448	1461	1448	633	1016	749.3	6.4	914.4	28-50.8	108	809.8	749.3	12.7	19.84	670	
	28	700	1549	1562	1549	684	1073	800.1	6.4	965.2	28-54	111.3	860.6	800.1	12.7	19.84	721	
	30	750	1651	1664	1651	735	1130	857.3	6.4	1022.4	28-54	114.3	917.4	857.25	12.7	19.84	772	
	32	800	1778	1794	1778	779	1194	914.4	6.4	1079.5	28-60.3	117.3	984.3	914.4	14.27	23.01	825	
	34	850	1930	1946	1930	830	1245	965.2	6.4	1130.3	28-60.3	120.7	1035.1	965.2	14.27	23.01	876	
	36	900	2083	2099	2083	874	1314	1022.4	6.4	1193.8	28-66.7	124	1092.2	1022.35	14.27	23.01	927	

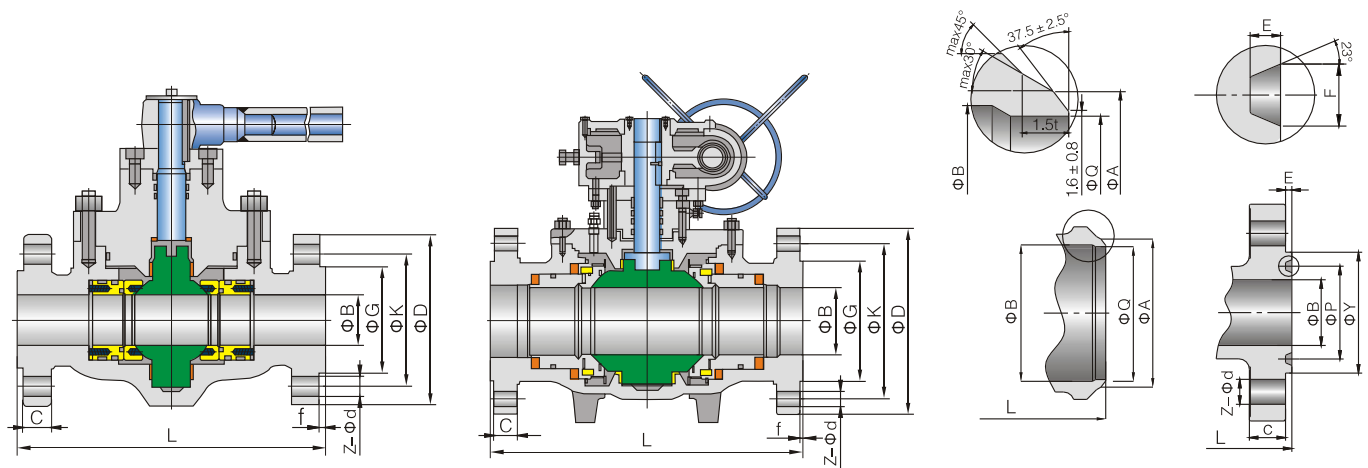
Full Port Top Entry Ball Valve



Main Connection Dimension

Class	Size		Dimensions(mm)															
	NPS	DN	L			B	D	G	f	K	Z-Φd	C	Groove				A	Q
			RF	RTJ	BW								Y	P	E	F		
900Lb	2	50	368	371	368	49	216	91.9	6.4	165.1	8-25.4	38.1	124	95.25	7.92	11.91	63	Asper client's requirement
	2 1/2	65	419	422	419	62	244	104.6	6.4	190.5	8-28.6	41.1	137	107.95	7.92	11.91	75	
	3	80	381	384	381	74	241	127	6.4	190.5	8-25.4	38.1	156	123.82	7.92	11.91	91	
	4	100	457	460	457	100	292	157.2	6.4	235	8-31.8	44.5	181	149.22	7.92	11.91	117	
	6	150	610	613	610	150	381	215.9	6.4	317.5	12-31.8	55.6	241	211.12	7.92	11.91	172	
	8	200	737	740	737	201	470	269.7	6.4	393.7	12-38.1	63.5	308	269.88	7.92	11.91	223	
	10	250	838	841	838	252	546	323.9	6.4	469.9	16-38.1	69.9	362	323.85	7.92	11.91	278	
	12	300	965	968	965	303	610	381	6.4	533.4	20-38.1	79.2	419	381	7.92	11.91	329	
	14	350	1029	1038	1029	322	641	412.8	6.4	558.8	20-41.3	85.9	467	419.1	11.13	16.66	362	
	16	400	1130	1140	1130	373	705	469.9	6.4	616	20-44.5	88.9	524	469.9	11.13	16.66	413	
	18	450	1219	1232	1219	423	787	533.4	6.4	685.8	20-50.8	101.6	594	533.4	12.7	19.84	464	
	20	500	1321	1334	1321	471	857	584.2	6.4	749.3	20-54	108	648	584.2	12.7	19.84	516	
	24	600	1549	1568	1549	570	1041	692.2	6.4	901.7	20-66.7	139.7	772	692.15	15.88	26.97	619	

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Main Connection Dimension

Class	Size		Dimensions(mm)															
	NPS	DN	L			B	D	G	f	K	Z-Φd	C	Groove				A	Q
			RF	RTJ	BW								Y	P	E	F		
1500Lb	2	50	368	371	368	49	216	91.9	6.4	165.1	8-25.4	38.1	124	95.25	7.92	11.91	63	Asper client's requirement
	2½	65	419	422	419	62	244	104.6	6.4	190.5	8-28.6	41.1	137	107.95	7.92	11.91	75	
	3	80	470	473	470	74	267	127	6.4	203.2	8-31.8	47.8	168	136.52	7.92	11.91	91	
	4	100	546	549	546	100	311	157.2	6.4	241.3	8-34.9	53.8	194	161.92	7.92	11.91	117	
	6	150	705	711	705	144	394	215.9	6.4	317.5	12-38.1	82.6	248	211.12	9.52	13.49	172	
	8	200	832	841	832	192	483	269.7	6.4	393.7	12-44.5	91.9	318	269.88	11.13	16.66	223	
	10	250	991	1000	991	239	584	323.9	6.4	482.6	12-50.8	108	371	323.85	11.13	16.66	278	
	12	300	1130	1146	1130	287	673	381	6.4	571.5	16-54	124	438	381	14.27	23.01	329	
	14	350	1257	1276	1257	315	749	412.8	6.4	635	16-60.3	133.4	489	419.1	15.88	26.97	362	
	16	400	1384	1407	1384	360	826	469.9	6.4	704.9	16-66.7	146.1	546	469.9	17.48	30.18	413	
	18	450	1537	1559	1537	406	914	533.4	6.4	774.7	16-73	162.1	613	533.4	17.48	30.18	464	
	20	500	1664	1686	1664	454	984	584.2	6.4	831.9	16-79.4	177.8	673	584.2	17.48	33.32	516	
	24	600	1943	1972	1943	546	1168	692.2	6.4	990.6	16-92	203.2	794	692.15	20.62	36.53	619	
2500Lb	2	50	451	454	451	42	235	91.9	6.4	171.5	8-28.6	50.8	133	101.6	7.92	11.91	63	Asper client's requirement
	2½	65	508	514	508	52	267	104.6	6.4	196.9	8-31.8	57.2	149	111.12	9.52	13.49	75	
	3	80	578	584	578	62	305	127	6.4	228.6	8-34.9	66.5	168	127	9.52	13.49	91	
	4	100	673	683	673	87	356	157.2	6.4	273.1	8-41.3	76.2	203	157.18	11.13	16.66	117	
	6	150	914	927	914	131	483	215.9	6.4	368.3	8-54	108	279	228.6	12.7	19.84	172	
	8	200	1022	1038	1022	179	552	269.7	6.4	438.2	12-54	127	340	279.4	14.27	23.01	223	
	10	250	1270	1292	1270	223	673	323.9	6.4	539.7	12-66.7	165.1	425	342.9	17.48	30.18	278	
	12	300	1422	1445	1422	265	762	381	6.4	619.2	12-73	184.2	495	406.4	17.48	33.32	329	